

OpenEvents V1: Large-Scale Benchmark Dataset for Multimodal Event Grounding

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The image depicts Donald Trump addressing the US Olympic team at a celebratory event likely held at the White House after the PyeongChang 2018 Winter Olympics. The photo captures a moment of national pride with Trump speaking to athletes dressed in their team uniforms. The image conveys a sense of unity and support for the team, but it is important to note the context of some athletes boycotting the event due to political disagreements with Trump.



In this somber photograph, a funeral procession winds through a park, carrying mourners to the final resting place of Richard Nieuwenhuizen, a 41-year-old volunteer linesman for SC Buitenboys. Richard Nieuwenhuizen tragically died after being beaten following an amateur football match, an event that has shaken the Dutch community, leading to the cancellation of 33,000 amateur games in his honor. His death has ignited a discussion about violence in football and the broader societal issues it reflects. This photograph captures the profound sorrow and unity felt by the community as they mourn the loss of a valued member.



The image captures a group of fans, led by Myriam Walter, heading to or from a court hearing in France. These fans are suing Wade Robson and James Safechuck, the men featured in the HBO documentary "Leaving Neverland," which detailed their allegations of sexual abuse against Michael Jackson. The lawsuit, which seeks symbolic damages and aims to defend Jackson's legacy.



The image captures a decisive moment from the UFC 207 women's bantamweight championship fight between Ronda Rousey and Amanda Nunes. This fight holds significant importance as Nunes defeated Rousey decisively, marking a turning point in women's MMA. In the photograph, Nunes is delivering a powerful punch to Rousey's face, visibly impacting her. The image vividly portrays the intensity and physicality of the fight, capturing the moment when Nunes' dominance shifted the course of the bout. This photo encapsulates a significant defeat for Rousey and the rise of Nunes as a dominant force in women's MMA.

Figure 1: Sample image-caption pairs in the proposed OpenEvents V1 dataset.

Abstract

We introduce OpenEvents V1, a large-scale benchmark dataset designed to advance event-centric vision-language understanding. Unlike conventional image captioning and retrieval datasets that focus on surface-level descriptions, OpenEvents V1 dataset emphasizes contextual and temporal grounding through three primary tasks: (1) generating rich, event-aware image captions, (2) retrieving event-relevant news articles from image queries, and (3) retrieving event-relevant images from narrative-style textual queries. The dataset comprises over 200,000 news articles and 400,000 associated images sourced from CNN and The Guardian,

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spanning diverse domains and time periods. We provide extensive baseline results and standardized evaluation protocols for all tasks. OpenEvents V1 establishes a robust foundation for developing multimodal AI systems capable of deep reasoning over complex real-world events. The dataset is publicly available at <https://ltnghia.github.io/eventa/openevents-v1>.

CCS Concepts

• Information systems → Information retrieval; • Applied computing → Media arts; • Computing methodologies → Computer vision; Machine learning.

Keywords

image captioning, image retrieval, multimodal dataset, real-world events, event-centric vision-language

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1 Introduction

Recent advances in vision-language models have significantly improved the ability of machines to describe and understand visual content through natural language. However, most existing image captioning benchmarks are grounded in synthetic or simplified scenarios, focusing primarily on object-level descriptions or everyday scenes [8]. These datasets and tasks fail to capture the complexity and contextual richness of real-world events, where understanding requires not just visual perception, but also temporal, social, and semantic context.

Understanding real-world events from images often requires more than just identifying objects or scenes, as it demands reasoning about what happened, who was involved, where and when it occurred, and why it matters. Such event understanding is critical in domains like journalism, historical archiving, disaster analysis, and media monitoring. However, current benchmarks lack the contextual depth needed to support these capabilities, as they rarely incorporate external knowledge or temporal dynamics. Moreover, captions generated in traditional settings are typically surface-level and visually grounded, offering limited insight into the broader narrative or consequences of the depicted moment.

To bridge this gap, we introduce the OpenEvents V1 dataset and establish a challenging, context-rich benchmark that mirrors how images are presented and interpreted in real-world settings, particularly within narrative news articles. By requiring models to jointly process both visual and textual content, this benchmark encourages the development of systems with deeper semantic grounding and event-level understanding. It further enables advancements in retrieval-augmented generation and multimodal reasoning.

Problem 1 – Event-enriched image captioning. Given an input image, models are required to retrieve relevant contextual information from an external article database and generate a detailed, context-aware caption (Figure 2a). This task encourages models to go beyond surface-level visual cues and incorporate real-world knowledge such as named entities, temporal context, and event outcomes. By grounding the caption in both visual content and textual evidence, the generated description reflects a more comprehensive understanding of the depicted event, supporting applications in visual journalism, archival, and automated media reporting.

Problem 2 – Event-based article retrieval. In practical deployments, an image captioning system may be provided with only the image, lacking the corresponding article. Because *event-enriched image captioning* requires both visual cues and contextual knowledge, the first step is to recover that missing context. To address this, we incorporate an article retrieval stage that locates the most relevant news article for the given image, ensuring the captioning model has access to the background information needed to produce accurate, event-aware descriptions (Figure 2b).

Problem 3 – Event-based image retrieval. Given a caption describing a real-world event, participants must retrieve the corresponding image from a large-scale database (Figure 2c). Unlike traditional keyword-matching systems, this task requires semantic alignment

between narrative-style text and event-centric visual representations. It emphasizes the ability of models to reason across modalities and handle diverse linguistic expressions of the same event. This capability is crucial for information retrieval in domains such as news curation, multimedia search engines, and digital content recommendation.

To support these tasks, OpenEvents V1 provides over 400,000 images and 200,000 articles collected from CNN and The Guardian, covering over a decade of diverse, high-impact events. Captions are generated using an improved version of the multi-stage VisChronos framework [10], which combines dense captioning, large language models, and contextual reasoning. In addition, we establish some baseline methods to encourage further research and benchmarking.

Our contributions are summarized as follows:

- We introduce OpenEvents V1, a large-scale benchmark for event-enriched image captioning and event-based image retrieval.
- We define and evaluate two event grounding tasks under a unified setting with standard metrics and some baseline models.
- We present empirical findings that reveal key challenges in handling real-world, context-rich vision-language data.
- We release OpenEvents V1 dataset at <https://ltnghia.github.io/eventa/openevents-v1>.

2 Related Work

Image-text datasets have been foundational to the progress of multimodal learning, particularly in retrieval tasks. Early datasets such as Flickr30K [7] provided high-quality annotated image-caption pairs, serving as standard benchmarks for evaluating cross-modal models. Building on this, large-scale datasets like WIT [16] introduced multilingual and diverse image-text pairs collected from Wikipedia, enabling broader pretraining and transfer to various downstream tasks.

As the field evolved, there has been increasing interest in domain-specific datasets that serve targeted applications. For example, Yuan et al. [22] proposed a biomedical image-text dataset for visual question answering, while Xu et al. [21] developed a dataset for agricultural disease diagnosis. These efforts demonstrate the importance of tailoring datasets to specific real-world contexts.

However, most existing datasets primarily focus on surface-level visual content, offering simple descriptions of what is visible in the image. They often lack deeper insights such as the names and attributes of entities, temporal aspects, and contextual outcomes [7] [22][21]. Even large-scale datasets like WIT [16] include broad coverage but provide limited depth regarding event-specific context and consequences. To date, no existing dataset is adequately suited to support retrieval tasks centered around real-world events, leaving a significant gap in multimodal research.

3 OpenEvents V1 Dataset

3.1 Dataset Construction

3.1.1 Data Source. To construct a comprehensive dataset for event-based image captioning, we collected a total of 202,803 news articles and 415,324 images from two major international news outlets: CNN and The Guardian.

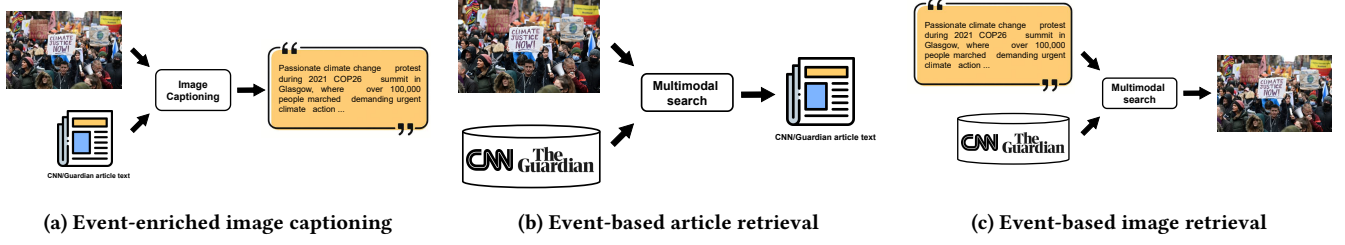


Figure 2: OpenEvents V1 dataset supports three tasks for event grounding.

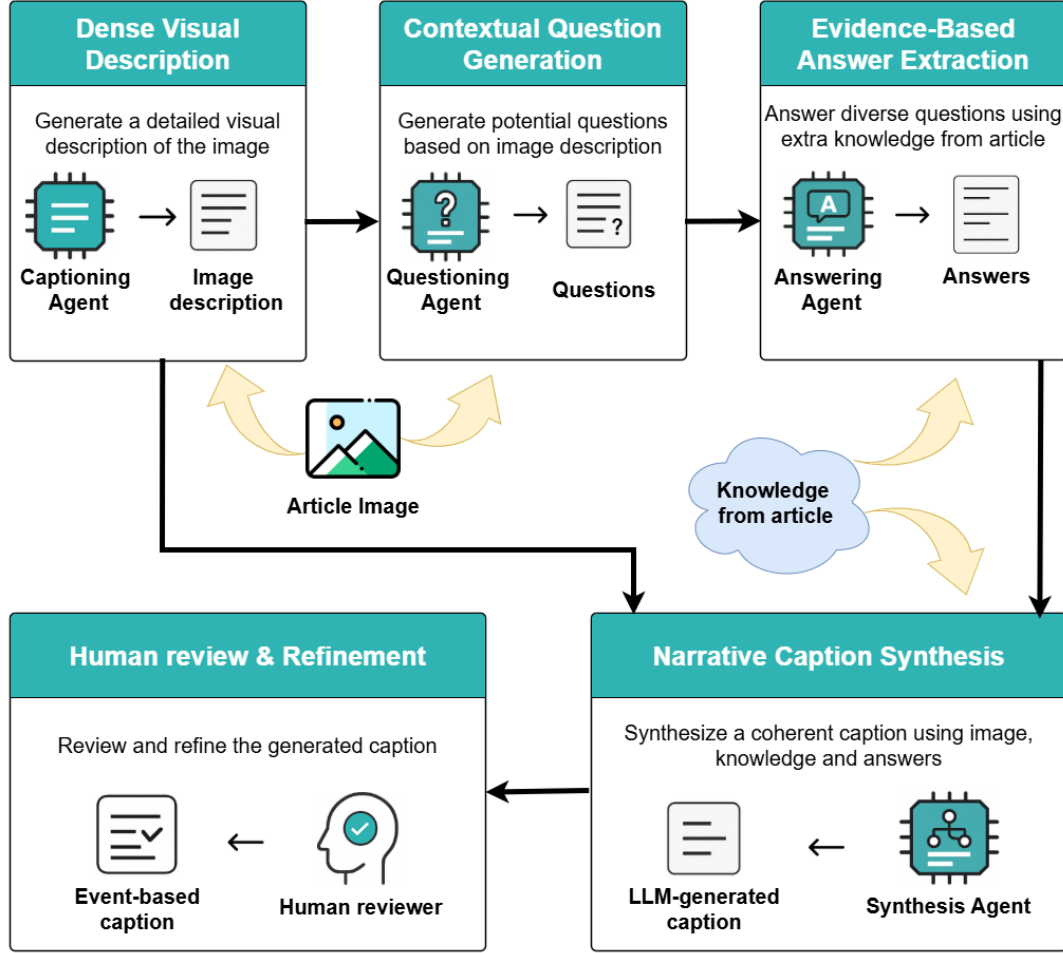


Figure 3: Human-agentic framework for dataset construction.

The CNN subset spans from 2011 to 2022, while The Guardian subset covers the period from 2019 to 2025. Both sources primarily focus on news and sports, but also contribute a rich mix of other domains such as health, politics, lifestyle, and culture as shown in Figures 4 and 5.

3.1.2 Human-Agentic Framework. To efficiently generate high-quality, contextually grounded captions at scale, we adopt a human-agentic

framework that combines automated language models and vision-language systems with final human validation. This five-stage pipeline builds on the original VisChronos [10], improves into modular agents, and incorporates an explicit human-in-the-loop step for final quality control. Examples of image-caption pairs generated by our framework are illustrated in Figure 1.

The process consists of the following stages: (1) Dense Visual Description, (2) Contextual Question Generation, (3) Evidence-Based

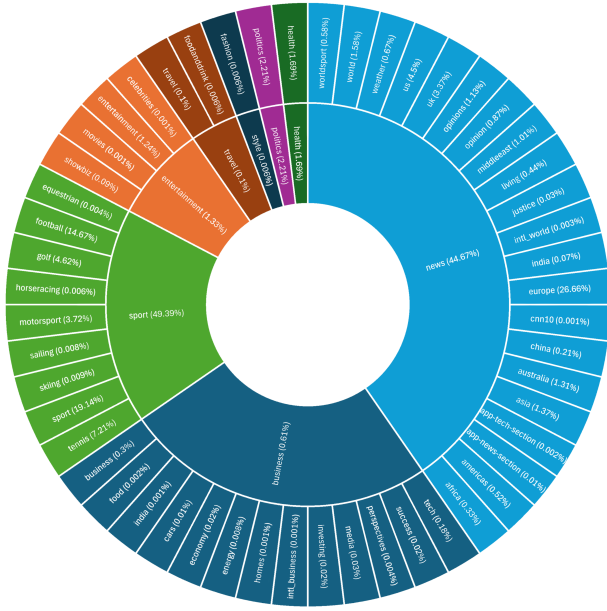


Figure 4: CNN article distribution across major content categories.

Answer Extraction, (4) Narrative Caption Synthesis, (5) Human Review and Refinement. This process can be seen as Figure 3

Step 1: Dense Visual Description. A dense captioning model is prompted to generate a detailed visual description of the input image, capturing salient objects, people, actions, interactions, and environmental context. This output serves as the foundation for subsequent semantic reasoning.

Compared to the original VisChronos framework [10], which relied on proprietary paid APIs, we adopt the open-source Molmo [2] model for improved accessibility and reproducibility. Furthermore, we refine the prompting strategy to encourage more descriptive and narrative-rich outputs, rather than simple enumerations of visual elements.

Step 2: Contextual Question Generation. Given the visual description, a language model generates a set of structured questions targeting the latent context behind the scene. These questions aim to uncover:

- Temporal and spatial setting (when, where)
- Main event and causality (what, why)
- Participants and roles (who)
- Outcomes, emotions, background, and implications

This step expands the captioning task from visual perception to contextual interpretation.

Step 3: Evidence-Based Answer Extraction. We provide each image with its associated article from the news corpus. The LLM is tasked with answering the generated questions by extracting sentence-level evidence from the article. Responses are only produced if the information is explicitly supported by the text. Otherwise, the model is instructed to return “no information.”



Figure 5: The Guardian article distribution across major content categories.

This stage ensures the factual grounding of generated content and prevents speculative generation.

Step 4: Narrative Caption Synthesis. The dense description and validated answers are synthesized into a coherent, paragraph-style caption. The caption aims to narrate the event depicted in the image, incorporating relevant names, dates, locations, outcomes, and contextual signals. The result is a rich, event-enriched caption that goes far beyond conventional object-centric descriptions.

Step 5: Human Review and Refinement. In the final stage, annotators assess the factual consistency with the article, correctness of named entities and temporal references, and overall readability. Captions with critical errors are edited or discarded. This procedure helps monitor quality and identify common failure patterns, contributing to the overall reliability of the dataset in practical applications such as retrieval, multimodal understanding, and media analysis.

3.2 Dataset Description

3.2.1 Statistic Information. The CNN subset of our dataset comprises 24,200 news articles and 89,596 images, spanning from 2011 to 2022. The Guardian subset includes a much larger collection, with 178,603 articles and 325,728 images, covering 2019 to 2025.

Figure 6 shows the annual distribution of articles and images. The number of samples rises sharply in 2019, then experiences a slight decline during 2020–2022, followed by a significant surge in 2023–2024, which represent the peak years in both article and image counts.

In terms of content, the dataset covers a wide range of topics. As shown in Figure 4 and Figure 5, just over half of the images belong to the news category, followed by significant portions in

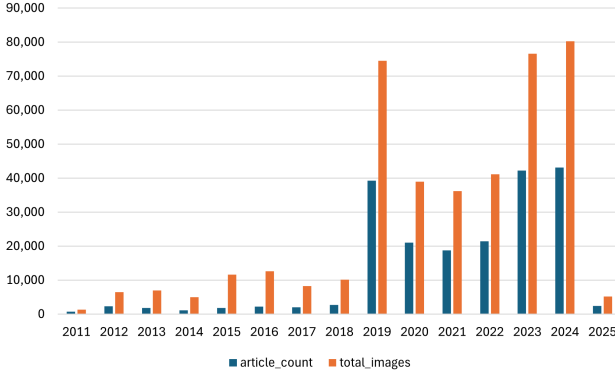


Figure 6: Distribution of articles and images by year (2011–2025).

sports. Despite this imbalance, the dataset still includes valuable long-tail sections such as arts, lifestyle politic and health offering diverse contextual grounding for caption generation.

3.2.2 Dataset Structure. The dataset is organized into four main subsets: train, public_test, private_test, and database. Each subset contains accompanying image files and structured annotations in JSON format.

- **Training Set:** This set contains 21,904 samples and is used for training captioning models.
- **Public Test Set:** This set comprises 6,000 samples, this set is openly available for evaluation and can be accessed by anyone, anytime.
- **Private Test Set:** This set consists of 4,000 samples and is reserved exclusively for the EVENTA 2025 challenge¹, with limited access available only during the competition period.
- **Database:** With 415,324 images and 202,803 textual descriptions of articles, this large-scale multimodal corpus serves as the reference set for retrieval-based tasks, enabling models to search and match relevant images given a query input.

All subsets except the database share a unified image-level JSON schema consisting of two fields. The first field, `image_id`, is a unique identifier assigned to each image. The second field, `caption`, provides the ground-truth description of the event depicted in the image.

In contrast, `database.json` operates at the article level and does not include captions directly. Instead, it provides context such as full article text and related image IDs. This structure supports a wide range of use cases, including robust training, zero-shot evaluation, and retrieval-augmented generation.

4 Benchmarking

We evaluate a set of lightweight baseline models for both two problems using the public test set of OpenEvents V1. Our goal is to provide reference points that are simple, reproducible, and suitable for benchmarking future models.

Each baseline is designed with minimal resource requirements, making them accessible and reproducible for all participants. We

¹<https://ltnghia.github.io/eventa/eventa-2025>

Table 1: Problem 1 – Captioning results on the public test set.

Method	CLIPScore	CIDEr	BLEU-4	METEOR
SmolVLM	0.4609	0.0044	0.0155	0.0789
SmolVLM + Article	0.5552	0.017	0.0229	0.0738
Qwen	0.5283	0.0282	0.0256	0.1320
Qwen + Article	0.5855	0.0565	0.0419	0.1383
Gemma	0.5945	0.0111	0.0243	0.1322
Gemma + Article	0.6634	0.0184	0.0341	0.1453

do not fine-tune on our dataset but use pre-trained models in with several different pipelines.

4.1 Problem 1: Event-Enriched Image Captioning

Unlike conventional image captioning tasks that rely solely on visual content, our formulation requires generating captions grounded in both the image and its corresponding article text. This setting reflects real-world scenarios, particularly in news media, where understanding the event depicted in an image requires contextual knowledge that goes beyond what is visually observable. The model must integrate visual semantics with textual evidence such as named entities, timelines, causes, and outcomes extracted from the associated article. This joint conditioning introduces unique challenges in cross-modal reasoning, factual consistency, and contextual relevance, making the task substantially more complex than traditional vision-language generation.

We experiment with several open-source vision-language models including SmolVLM [9], Qwen2.5-3B [5], and Gemma-3-4b [17], following two main pipelines: (1) *Image-only*, where the model generates captions solely from the image; and (2) *Image + Article*, where the model is additionally provided with the corresponding article text. The latter approach aims to simulate real-world settings where understanding an event often requires both visual perception and contextual information.

We evaluate the ability of models to generate context-rich captions that go beyond surface-level visual description by incorporating external event knowledge. To assess caption quality, we report standard captioning metrics including CLIPScore [4], CIDEr [20], BLEU-4 [12] and METEOR [6], which measure fluency, descriptiveness, and semantic alignment with reference captions.

As shown in Table 1, models in the *Image-only* pipeline (e.g., SmolVLM, Gwenn) consistently underperform across all metrics when compared to their *Image + Article* counterparts. This performance gap underscores the importance of contextual grounding. Because the ground-truth captions contain event-specific details such as named entities, temporal markers, causes, and outcomes, they cannot be accurately reconstructed from visual content alone. Incorporating article-level textual evidence is therefore crucial for generating factually grounded and semantically enriched descriptions, highlighting the value of retrieval-augmented generation in real-world image captioning scenarios.

4.2 Problem 2: Event-Based Article Retrieval

We evaluate CLIP[13] and OpenCLIP[1] on four ranking-based metrics, such as mean Average Precision (mAP), Normalized Discounted Cumulative Gain (NDCG), Nearest Neighbor accuracy

Table 2: Problem 2 – Article retrieval results on the public test set.

Method	mAP	NDCG	NN	AUC
CLIP	0.3504	0.4594	0.23	0.0575
OPEN CLIP	0.3141	0.4249	0.2016	0.0487

Table 3: Problem 3 – Image retrieval results on the public test set.

No.	Method	mAP	NDCG	NN	AUC
1	CLIP	0.2467	0.3407	0.1586	0.0302
2	OPEN CLIP	0.1845	0.2703	0.1845	0.0185
3	SBERT + Flan T5	0.2134	0.2837	0.1376	0.0220
4	SBERT + Bart	0.2840	0.3628	0.1863	0.0372
5	SBERT + Pegasus	0.2868	0.3665	0.1930	0.0362
6	SBERT + Flan T5 + CLIP	0.2795	0.3408	0.1986	0.0303
7	SBERT + Bart + CLIP	0.3232	0.3978	0.2226	0.0436
8	SBERT + Pegasus + CLIP	0.3216	0.3986	0.2173	0.0450

(NN), and Area Under the Precision–Recall Curve (AUC). Results in Table 2 show a consistent advantage for CLIP across all measures, suggesting superior alignment with the event-centric properties of OpenEvents V1.

This performance difference may be attributed to differences in training data. CLIP, the model released by OpenAI, was trained on a carefully curated dataset with strong image–text correspondence. OpenCLIP, while architecturally similar, was trained on larger but less targeted public datasets, such as LAION [15], DataComp [3], CommonPool [11], and YFCC100M [18], that may not reflect the style and content of journalistic imagery. This difference in data curation likely explains CLIP’s stronger performance, reinforcing the broader insight that domain alignment in training data is critical for high retrieval accuracy in event-focused tasks.

4.3 Problem 3: Event-Based Image Retrieval

We evaluate the capability of models to retrieve the correct image from a large-scale visual database given a caption that narrates a real-world event. Unlike conventional retrieval tasks, where captions typically describe only the visual content, our setting presents a greater challenge: the caption is generated based on both the image and its associated article text. As a result, the descriptions often reflect deeper event semantics, such as named entities, temporal references, and contextual outcomes that may not be visually evident. This makes direct retrieval significantly harder, as models must align complex, narrative-style captions with visual content that is only partially represented in the text.

Table 3 summarizes the results on the public test set. Article-guided retrieval consistently outperforms direct image-based approaches across all metrics, highlighting the value of contextual grounding in event-centric search.

The first two baselines (CLIP [13] and OpenCLIP [1]) follow a direct approach that retrieves images purely based on query-to-image similarity. OpenCLIP [1] performs worse than CLIP [13], possibly because it was further trained on public datasets that differ from the news domain. These two methods yield relatively lower

results, indicating limited capacity to align complex, event-rich queries with image content in the absence of external context.

In contrast, baselines 3 to 5 incorporate an intermediate article retrieval stage. In this approach, the system first retrieves a relevant article and then selects associated images based on their original order within that article. To facilitate this process, we employ pre-trained language models such as Flan-T5, Pegasus, and BART to generate concise textual summaries for each article. During inference, the input caption is embedded using SBERT [14] and matched against these summaries to identify the most semantically relevant article. This article-guided strategy yields substantial improvements in retrieval performance. Notably, the SBERT + Pegasus configuration achieves 0.2868 mAP and 0.3665 NDCG, outperforming all direct retrieval baselines.

Further improvements are observed in Baselines 6 and 7, which re-rank the top candidate images from the retrieved article using CLIP similarity. These hybrid approaches (e.g., SBERT + Bart + CLIP and SBERT + Pegasus + CLIP) achieve the highest performance across all metrics, with mAP scores exceeding 0.32 and NN accuracy above 0.22. This demonstrates the benefit of combining article-level semantic filtering with fine-grained visual similarity re-ranking.

Overall, the results highlight the importance of contextual grounding: retrieval methods that leverage both textual semantics and visual similarity consistently outperform those relying solely on direct query-to-image matching.

5 Conclusion

In this paper, we introduced OpenEvents V1, a large-scale benchmark designed to advance research in event-enriched image captioning and event-based image retrieval. To ensure high-quality outputs for event-enriched image captioning, we proposed a five-stage human–agentic framework that integrates vision–language models with article grounding and final human verification. This framework supports three benchmark tasks: event-enriched image captioning, event-based article retrieval, and event-based image retrieval. Baseline results for all tasks demonstrate the challenges and opportunities presented by this dataset, filling a critical gap in multimodal research through large-scale, real-world supervision enriched with contextual information. Our benchmark and baselines have been made publicly available to facilitate reproducibility and comparison. This dataset also has been utilized in the ACM Multimedia Grand Challenge on Event-Enriched Image Analysis (EVENTA’25)[19] to encourage the development of context-aware multimodal AI systems.

The OpenEvents V1 dataset can support a wide spectrum of real-world and research-oriented applications. Its dual focus on captioning and retrieval makes it an ideal testbed for evaluating retrieval-augmented generation systems and hybrid LLM architectures. The integrated article database provides a robust foundation for applications where external factual grounding is essential, enabling adaptations for visual question answering, entity extraction, and multimodal reasoning.

Future work will focus on expanding the benchmark with multilingual content, multimodal event graphs, and video-based event understanding to capture richer temporal dynamics.

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